Structural proxies explored for amide synthesis

Full historical inventory, grouped by category; subsequent matched comparisons included below.

Category	Full proxy list (closely related variants grouped)	Fields
Selected: proximity-weighted attack angle	$A = \sum_i(\theta_i/d_i) \,/\, \sum_i(1/d_i)$; report harmonic distance $d_H = N/\sum_i(1/d_i)$ alongside it. Larger A ranks higher.	Selected
Activation geometry / local environment	Mean distance-gated in-line angle; in-line fraction among gated poses; near-attack fraction; reactive-group SASA; Mg–P and Mg–O distances; Mg coordination; basic and acidic residue counts within 6 Å; gated-pose count.	10
Geometry summaries	Boltz-2 minimum approach distance, gated mean angle, near-attack fraction, and total pose count.	4
Amine attack (step 2)	Minimum amine-N–acyl-C distance; near-attack fraction; gated Bürgi–Dunitz angle; gated-pose count.	4
Mechanistic hypothesis battery	Reactive-carboxylate polar contacts; distal-carboxylate SASA; acceptor-channel clearance; distal-minus-reactive SASA; minimum amine approach; amine approach fraction within 6 Å; adenine burial; acyl-carbonyl polar contacts; step-1 gated and step-2 bound pose counts.	10
Preorganization / pose consistency	Reactive-carboxylate centroid RMSF; attack-direction angular spread; bidentate-clamp fraction; mean attack angle; gated-pose count.	5
Intermediate exposure / pocket engagement	SASA of distal carboxyl carbon, acyl carbon, carbonyl oxygen, and reactive center; median intermediate-to-protein distance.	5
Substrate access / tunnel geometry	Open-channel length at clash cutoffs 2.5, 2.7, and 3.0 Å; tunnel enclosure under tight, medium, and loose settings; solvent-exit fraction; gated-pose count.	8
Amine-subsite charge environment	Acidic-oxygen counts, distance to a nearby candidate base, and net charge near probes placed 6, 7, 8, and 9 Å beyond the distal carboxylate (6 Å charge-counting shell); gated-pose count.	13
Electrostatic potential / electric field	Electrostatic potential and projected local electric field at ATP P_α and the acyl carbonyl; ensemble mean and standard deviation for each.	8
Protonation / acid–base environment	For steps 1 and 2: minimum histidine $\lvert pK_a - 7\rvert$, minimum lysine pK_a, and ionizable-residue count (each with ensemble mean and standard deviation); processed-pose counts.	14
Metal coordination	Mean/median Mg–P distances; minimum/mean Mg–O distances; coordinated fraction; Mg-bridging fractions at 2.2, 2.5, 3.0, and 3.5 Å; median donor-O–P approach distance.	10
Interface-confidence baselines	Directional enzyme→substrate and substrate→enzyme ipTM (ensemble means and maxima); symmetric mean; minimum and maximum of the directional means.	7
Matched follow-up: geometry variants	Angle summaries (mean, best, best-five mean, 90th percentile); distance-only score, minimum distance, and nearby fraction; distance-gated and gated-weighted angles (including minimum-five-pose variants); inverse-distance-weighted angle/alignment, angle-over-distance and alignment-over-distance (including positive alignment); raw weighted angle; hard/soft near-attack fractions; joint distance–angle compatibility (mean, best, best-five mean, weighted 90th percentile); regularized/unregularized score; ideal-position reorganization distance (best, best-five mean, mean); interface confidence.	29 formulas
Matched follow-up: consistency, contacts, combinations	Attack-direction consistency; donor-centroid, ATP P_α, and relative donor–P_α RMSF; two-oxygen polar-contact fraction; severe-overlap score; historical score multiplied by direction consistency, donor rigidity, relative rigidity, or polar-contact fraction.	10 formulas

Inventory: 98 historical exported fields = 88 distinct nonconstant definitions + 5 duplicate exports + 5 constant/undefined fields. The 39 matched formulas overlap with the historical inventory; these counts are not additive. Counts include sampling diagnostics and formula variants.

Provenance: Imported amide discovery run and associated structural-screening artifacts; historical field inventory (2026-10-11), matched comparisons, and current attack-angle policy. Exact identifiers and source hashes accompany this PDF in `sources.json`.

Interpretation: An inventory of explored descriptors, not evidence that each predicts activity. SASA = solvent-accessible surface area; RMSF here measures spread across generated poses. Polar contacts are geometric proxies.